

Bachelor of Science in Civil Engineering

Year of study: 2nd Year; 2nd Semester

Course Title: CIVIL ENGINEERING DRAWING

Course Code: FCC 3242

LECTURE FOUR
Development of Surfaces

Lecture Outline

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- 4.2 Learning Outcomes
- 4.3 Introduction to surface development
- 4.4 Procedure of developing a surface
- 4.5 Ability to develop an object
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4.1 Introduction

Welcome to the fourth lecture in which we will discuss the “opening out” of hollow engineering objects followed by drawing the resultant shape. In this lecture, we begin with simple developable objects and apply the four types of development to produce the desired drawings. The importance of an accurate development drawing cannot be over-emphasized since it is from this drawing that the material making the hollow object is made. This material, so defined by the development drawing, is either folded or rolled to make the intended object.

4.2 Learning Outcomes

At the end of this lecture, you should be able to:

- a) Draw the orthographic views and surface development drawings of hollow engineering objects.
- b) Meet the quality expectations for the drawings produced

4.3 Introduction to surface development

Consider the following objects:

- a) Household items such as: cartons, metal boxes, tin cans, funnels,
- b) Engineering objects such as:
 - i. Furnace chutes, ducts, roof gutters etc.
 - ii. Steel pipes (such as the ones shown below) and corresponding elbows/bends,



These items are made from flat sheet material which when folded, formed or rolled will take the shape of the object.

Definition

Development is the process of unfolding the surfaces of an object and spreading them on a flat plane.

4.4 Procedure of surface development

1. Drawing the regular orthographic views of the object as discussed in previous topics.

Then

2. Draw the development of the object to show the complete surfaces laid out in a flat plane.

Identification of corners is once again important in this procedure

4.5 Ability to develop an object

An object is said to be developable if a thin sheet of flexible material e.g. paper can be wrapped smoothly about the object.

a) Developable

Objects with plane or flat surfaces or single curved are developable

b) Undevelopable

Objects which are double curved or warped e.g. a sphere; approximate methods must be used to develop these objects (think of a sphere as an object with infinite number of regular polygons)

4.6 Types of developments

There are four types of development in common use, namely:

- a) Straight-line development of prism surfaces
- b) Radial line development of pyramid surfaces
- c) Radial line development for conical surfaces
- d) Parallel line development of cylindrical surfaces

4.7 Straight line development

This is the development of an object that has flat surfaces such as a prism.

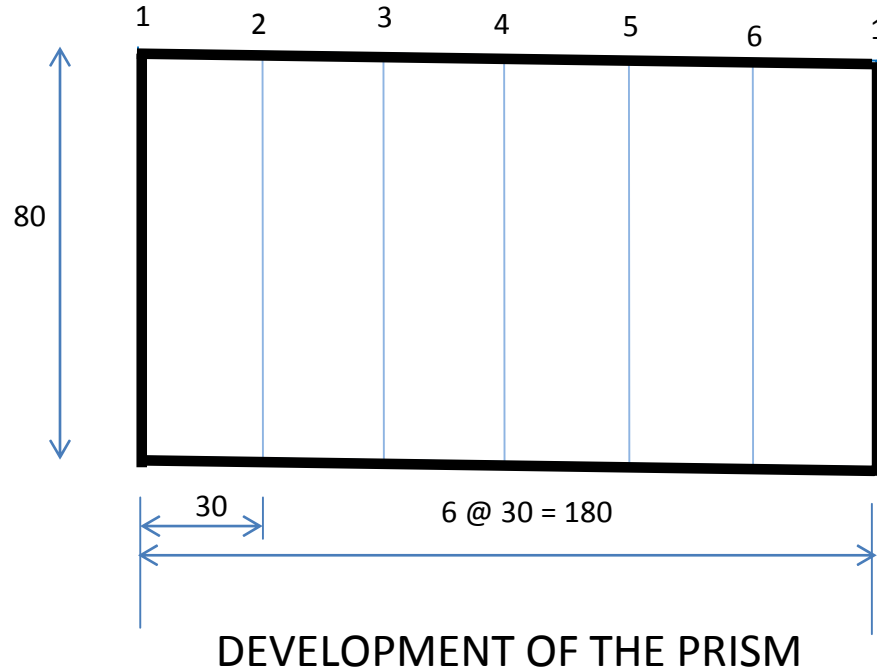
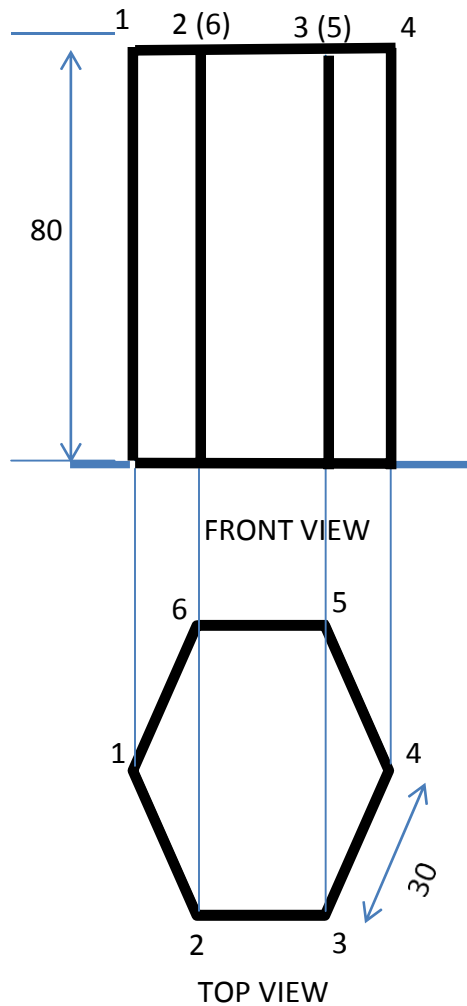
- a) The *true size* of each side of the object is known
- b) Sides can be laid out in successive order.

The example below shows the development of a hexagonal prism (i) placed with one side parallel to the FPP) and (ii) after it has been truncated using a cutting plane inclined at an angle Θ to the horizontal but which is normal to the FPP. The cutting plane passes at a point on the axis which is 50mm above the base of the prism.

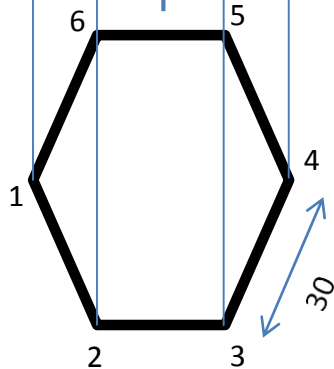
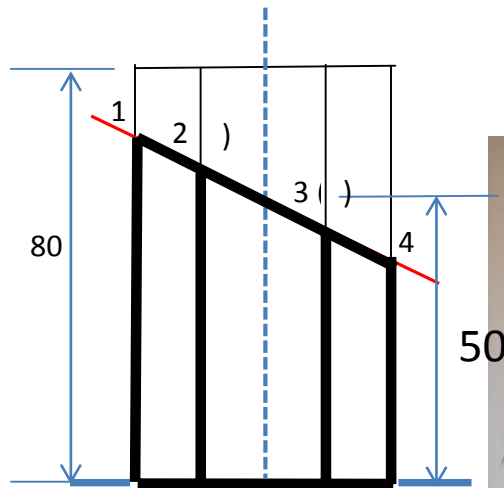
Note:

- a) The fold lines are shown as thin, visible continuous lines.
- b) All lines for each surface are straight.

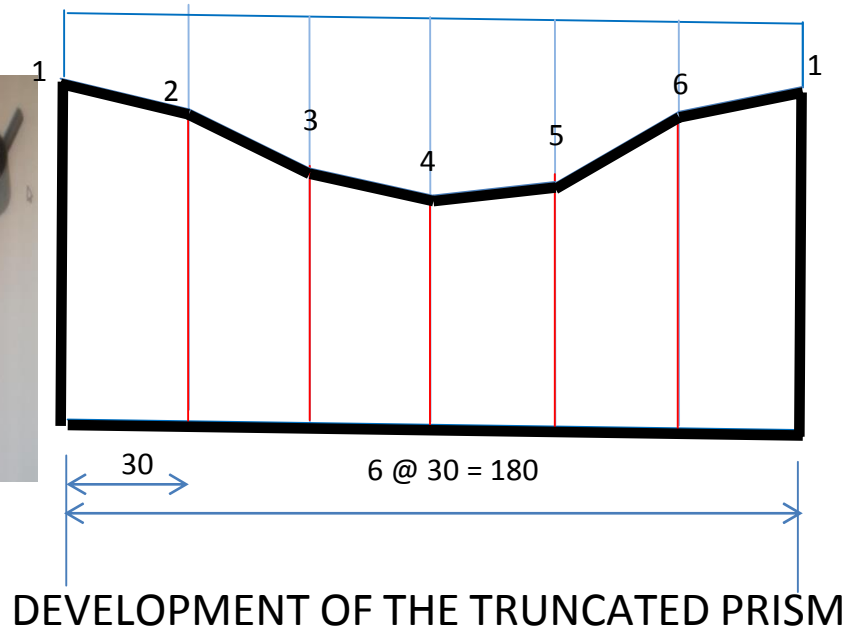
Full hexagonal prism - animate



Truncated hexagonal prism - animate



TOP VIEW



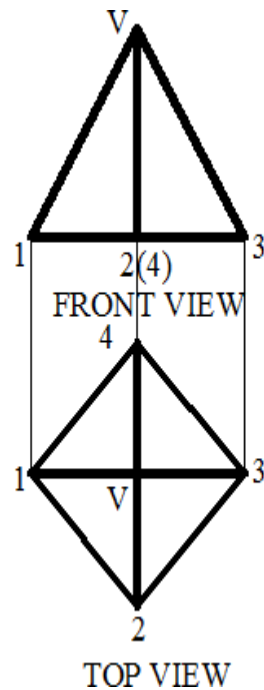
4.7 Radial line development of pyramid surfaces

Radial line development of a pyramid is illustrated here with the help of a right square pyramid arranged in various positions. Remember that for a right pyramid, all lateral edges (from vertex to the base) are equal.

Position 1

The pyramid is positioned in such a way that at least two lateral edges are in true length.

Looking at the Front and Top Views of the pyramid in this orientation, the following observations can be made:



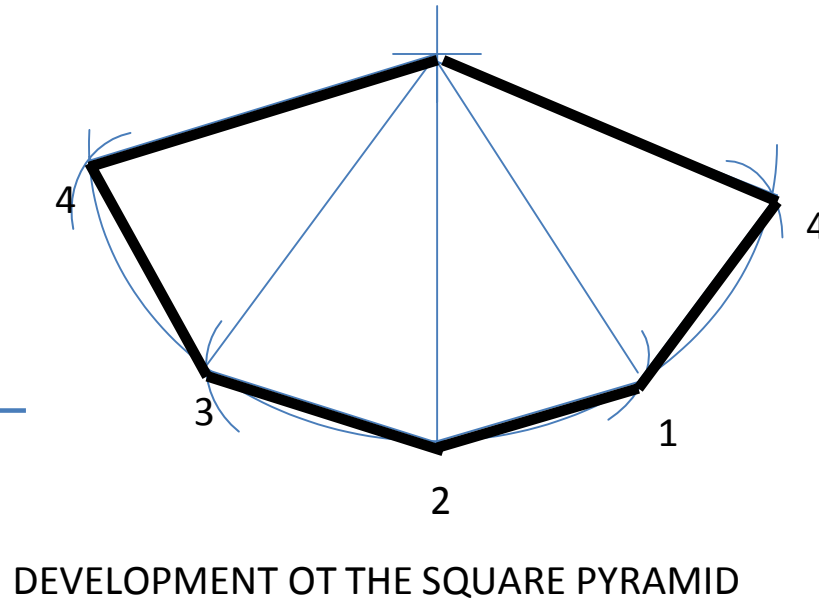
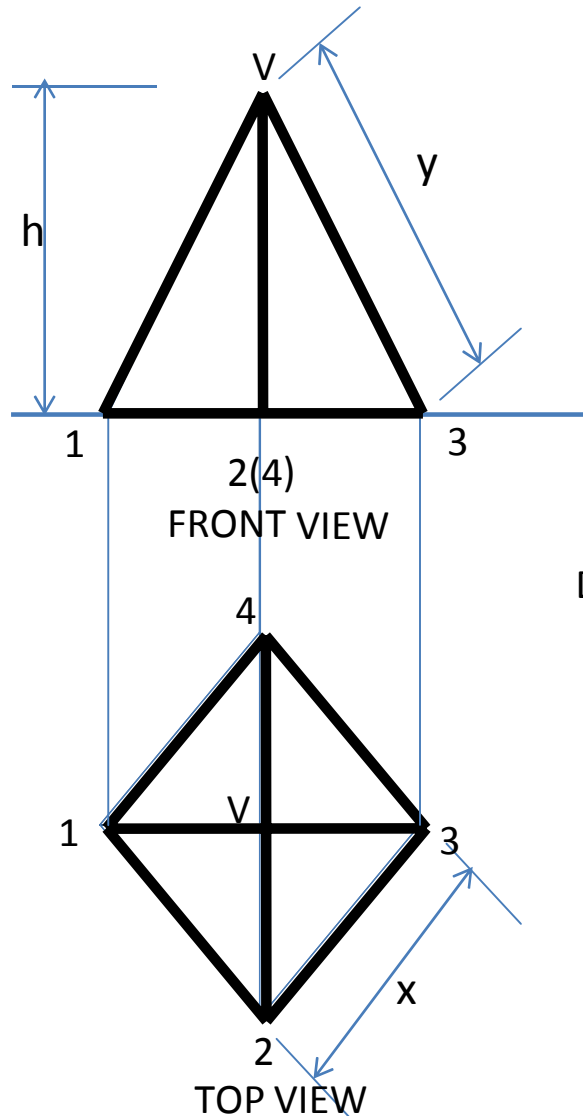
View	True lengths	Untrue or projected lengths
Front View	1 – V; 3 – V Thus can be used for development	2 – V; 4 – V Cannot be used for development
Top view	1 – 4; 4 – 3; 3 – 2; 2 – 1 Thus can be used for development	1 – V; 2 – V; 3 – V; 4 – V Cannot be used for development

Construction of the development

Use ***only the true lengths*** in the construction of the development

1. With V as centre, draw an arc of radius V – 1 or V – 3 (true edge length)
2. Drop a perpendicular from V to cut the arc at 2 (you have chosen to slit the pyramid through V – 4 and assumed the base is hollow)
3. With radius equal to the size of the side of the square base, step off distance 2 – 3, 3 – 4 and 2 – 1, 1 – 4 around the arc, starting at 2
4. Join these points with straight lines
5. Then connect these points to V

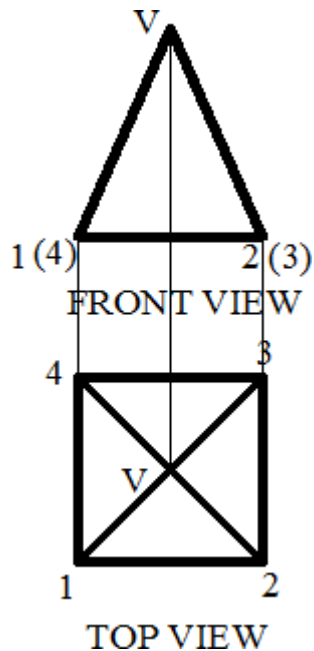
Radial line development of a square pyramid - animate



Position 2

The pyramid is positioned in such a way that **none** of the lateral edges are in true length.

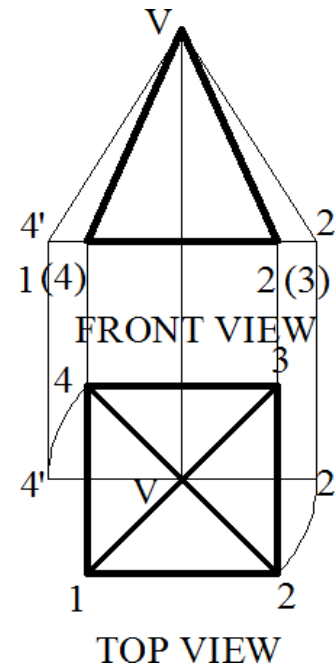
Looking at the Front and Top Views of the pyramid in this orientation, the following observations can be made:



View	True lengths	Untrue or projected lengths
Front View	1 – 2; (4) – (3) Thus can be used for development	1 – V; 2 – V; 3 – V; 4 – V Cannot be used for development
Top view	1 – 4; 4 – 3; 3 – 2; 2 – 1 Thus can be used for development	1 – V; 2 – V; 3 – V; 4 – V Cannot be used for development

Construction of the development

1. Since none of the lateral edges has a true length in this position, rotate the top view (anti-clockwise e.g.) such that the vertical plane containing 4-V-2 (or 3-V-1 clockwise) is parallel to FPP as shown in the Top View on the right. Project the new positions of corners 2 and 4 (i.e. 2' and 4') to the Front View.
2. Either V-4' or V-2' is the true length of the lateral V-4 and the others.
3. Using this as the radius, proceed as in previous position.



Remember the octahedron in Tutorial 1(d) !!

Truncated pyramid in Position 2

To obtain the true length of the lateral edge (V-4' or V-2') the same procedure is followed as before but only the portions of the lateral edges that remain after truncation are required for the development.

Look at the slide below

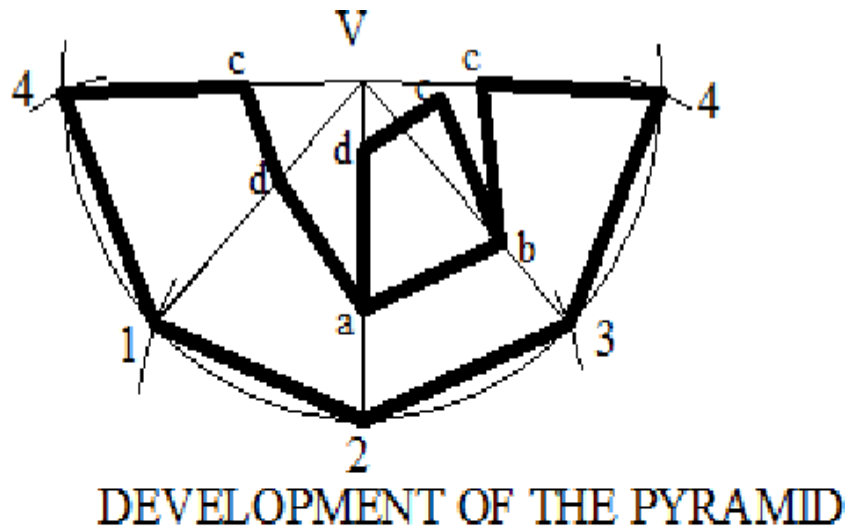
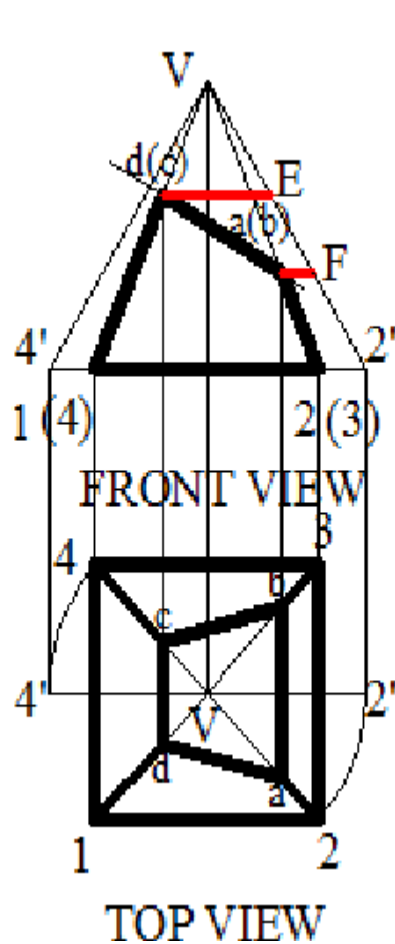
In the front view, draw a horizontal line from d or (c) to cut the lateral V-2' at E. Do likewise from a or (b) to F.

Note the following:

- i. The true length of portion 2-a or 3-b is 2'-F
- ii. The true length of portion 1-d or 4-c is 2'-E

These lengths are the ones to be transferred to the development in addition to the other true lengths.

The flap a-b-c-d is also optionally added e.g. if given that the top is covered.



Note

The same procedure for obtaining true lengths can be used for other objects such as Oblique Pyramids etc.

4.9 Radial line development for conical surfaces

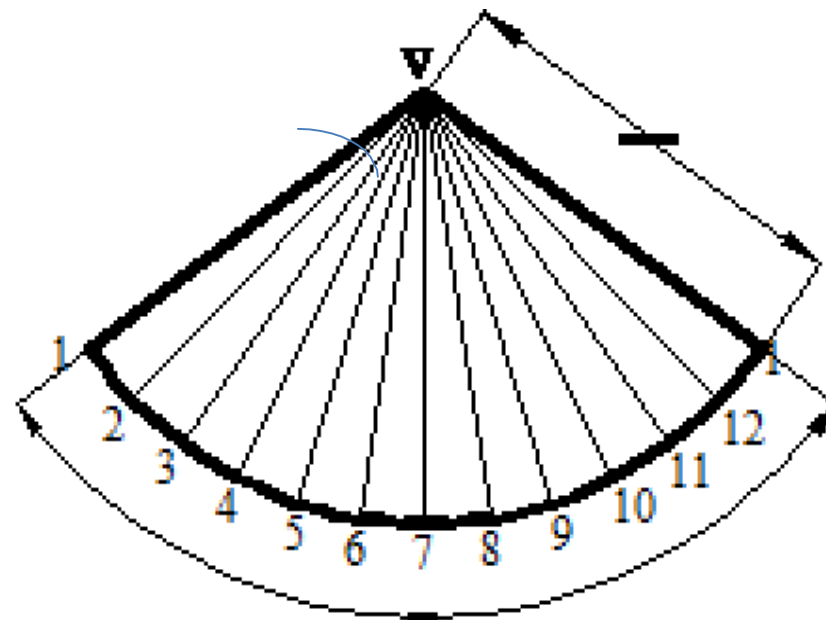
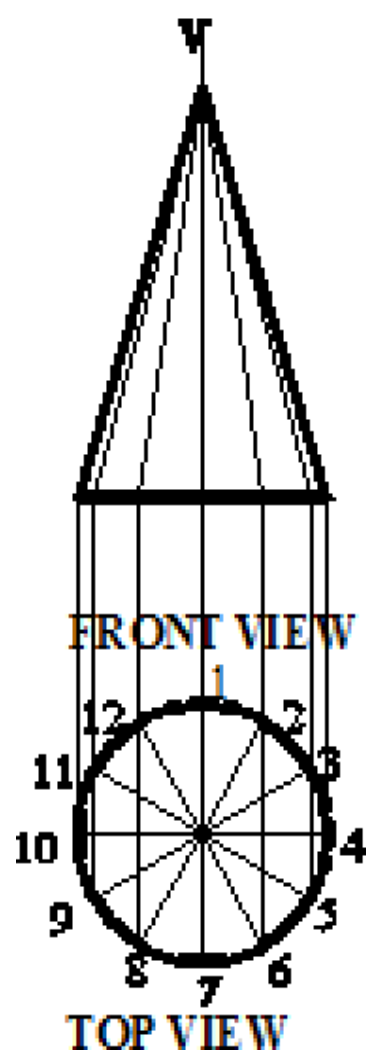
The following lengths are needed to develop a cone:

- a) Slant length
- b) Circumference of base

The development of a cone is a **sector** of a circle of radius equal to the slant length.

Following is an example of a right cone (i.e. one whose vertex is directly above the centre of base) and the truncated piece of the same.

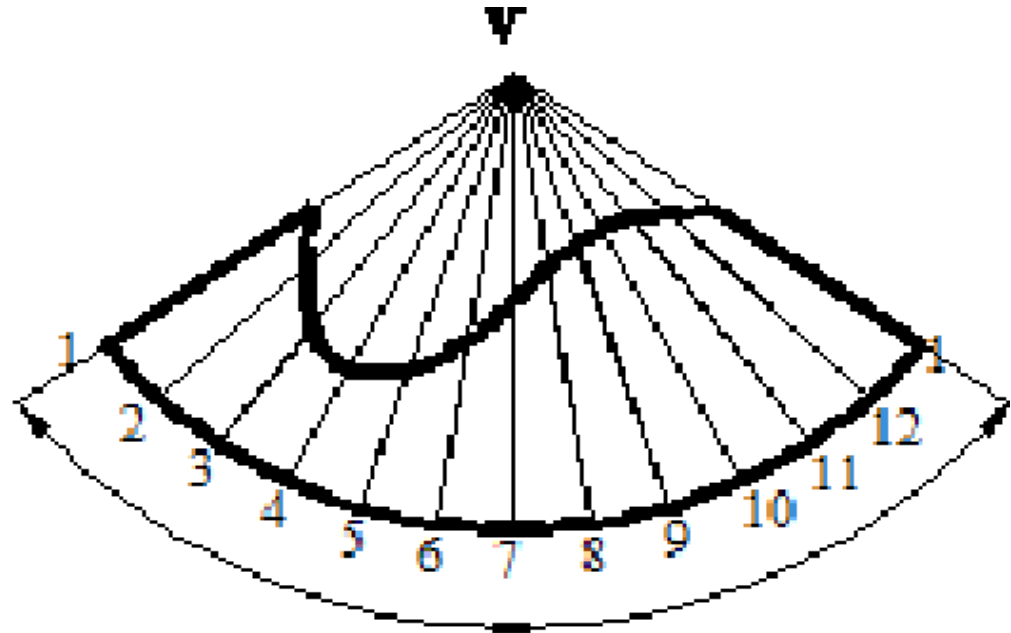
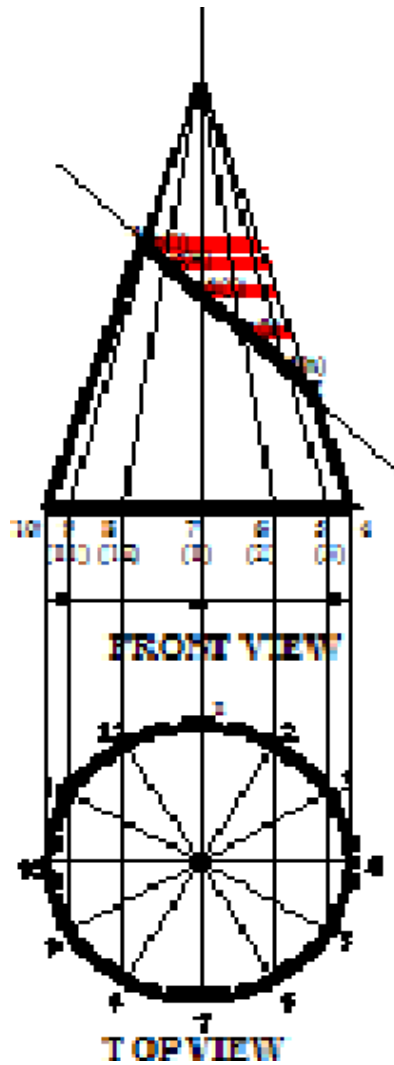
Note: The procedure of getting true lengths for development has been explained before and can also be used for oblique cones.



DEVELOPMENT OF THE



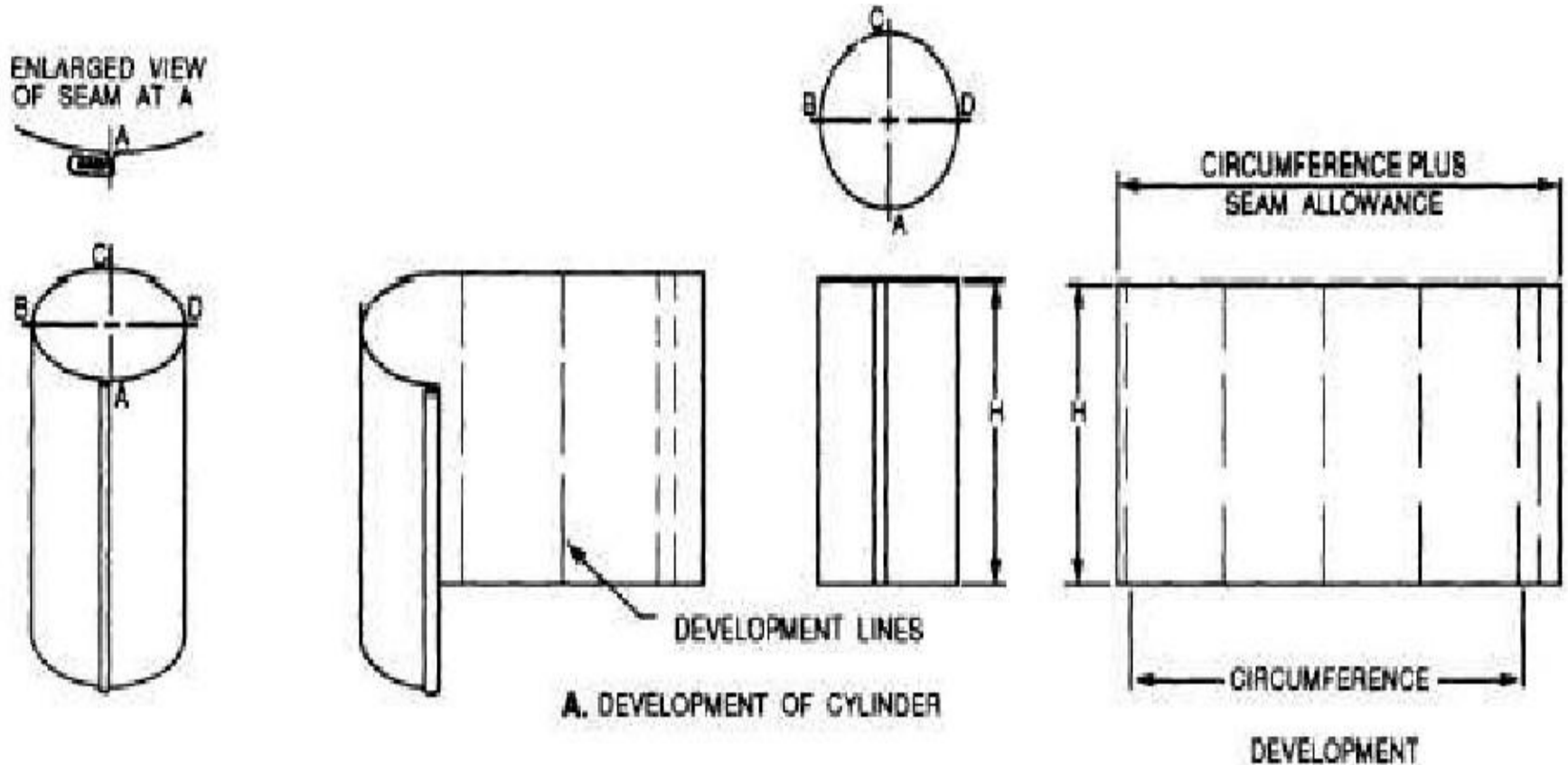
Truncated Cone



DEVELOPMENT OF THE
TRUNCATED CONE

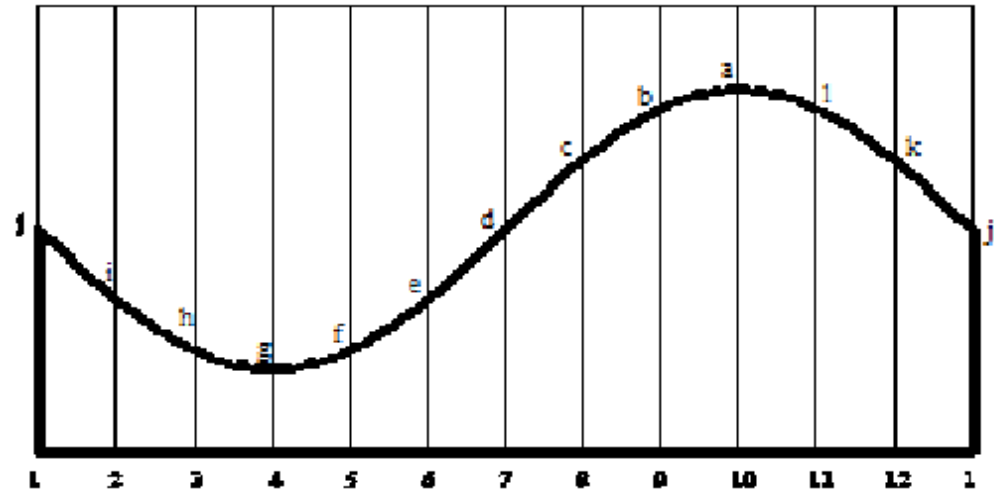
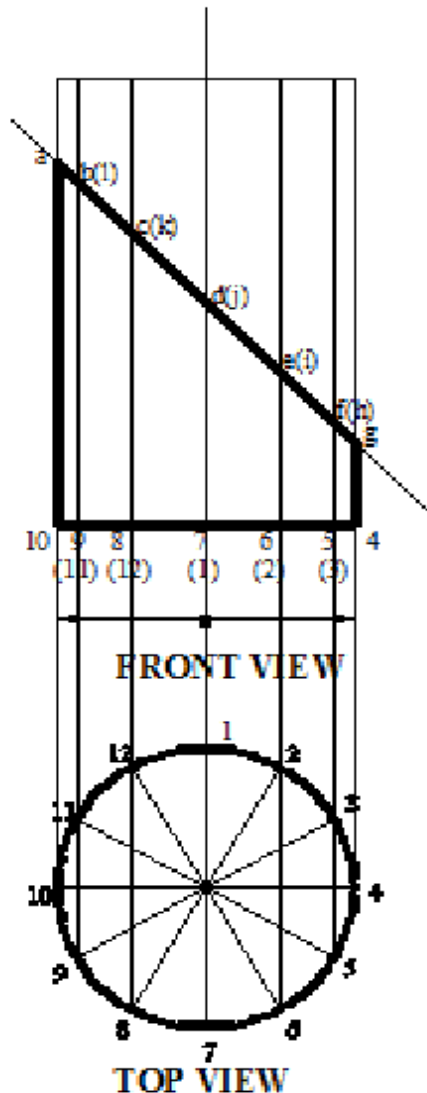
4.10 Parallel line development of cylindrical surfaces

Cylindrical surfaces are developable since they are made of single curved surfaces of constant radius.



Truncated cylinder

All distances are parallel to the FPP and therefore true lengths and can be used directly in the development.



DEVELOPMENT OF THE TRUNCATED CYLINDER

4.11 Summary

We have come to the end of lecture four. This lecture started by defining what it means to develop a hollow object. We then discussed various methods of developing surfaces of common, simple objects and the resultant development drawings.

4.12 Self - Assessment

Roll A4 paper (210 x 297) into:

(a) Cylinder

(b) Cone

4.13 References and further reading

References for further reading were listed in Lecture one. You are free to look at any of them for the purpose of this lecture.